

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

03

10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS

A special camera is used for underwater ocean research. The camera is at a depth of 39 fathoms. What is the camera's depth in feet? 1 fathom = 6 feet

- A. 234
- B. 117
- C. 45
- D. 7

At a large high school, 300 students were selected at random and were asked in a survey about a menu change in the school cafeteria. All 300 students completed the survey. It was estimated that 38% of the students were in support of a menu change, with a margin of error of 5.5%. Which of the following is the best interpretation of the survey results?

- A. The percent of the students at the school who support a menu change is 38%.
- B. The percent of the students at the school who support a menu change is greater than 38%.
- C. Plausible values of the percent of the students at the school who support a menu change are between 32.5% and 43.5%.
- D. Plausible values of the number of the students at the school who support a menu change are between 295 and 305.

Q3

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

Out of 300 seeds that were planted, 80% sprouted. How many of these seeds sprouted?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

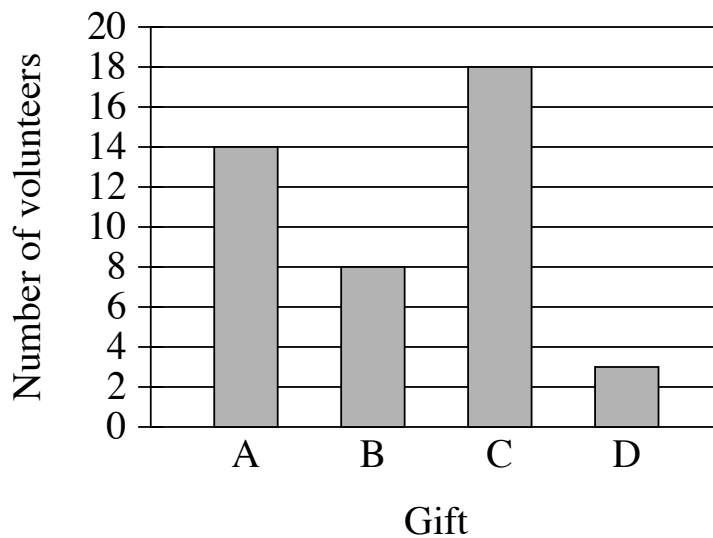
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A survey taken by 1,000 students at a school asked whether they played school sports. The table below summarizes all 1,000 responses from the students surveyed.

	Males	Females
Play a school sport	312	220
Do not play a school sport	?	216

How many of the males surveyed responded that they do not play a school sport?

- A. 109
- B. 252
- C. 468
- D. 688

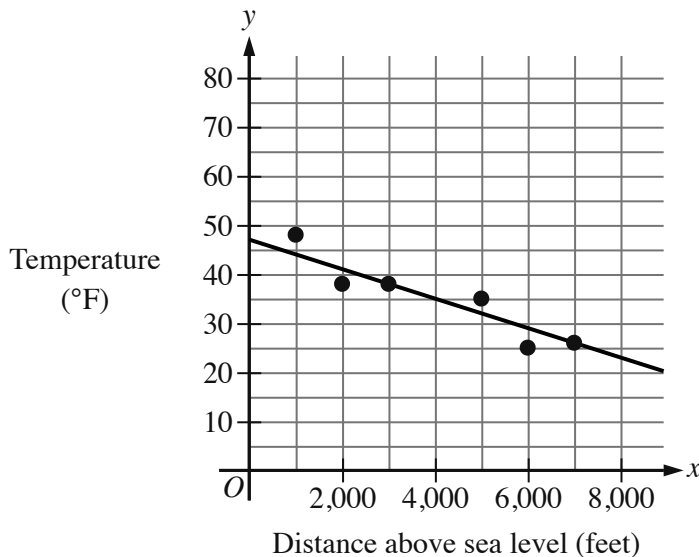


- The Number of volunteers data for the 4 bars are as follows:
 - A: 14
 - B: 8
 - C: 18
 - D: 3

In April, there were 43 volunteers in a cleanup project. Each volunteer was asked to choose a small gift labeled A, B, C, or D. The bar graph shows the number of volunteers who chose each gift. How many volunteers chose gift C?

- A. 3
- B. 8
- C. 14
- D. 18

The scatterplot shows the temperature, in degrees Fahrenheit $^{\circ}\text{F}$, and the distance above sea level, in feet, measured at 6 locations on Mount Jefferson. A line of best fit is also shown.



- The scatterplot has 6 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
 - The line of best fit slants down from left to right.
 - The line of best fit passes through the following coordinates:
 - (0 comma 47)
 - (4,000 comma 35)
 - (7,000 comma 26)

At a distance of 4,000 feet above sea level, what is the temperature, in $^{\circ}\text{F}$, predicted by the line of best fit?

A. 47

B. 35

C. 25

D. 0

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

If $t = 4u$, which of the following is equivalent to $2t$?

- A. $8u$
- B. $2u$
- C. u
- D. $\frac{1}{2}u$

A random sample of 50 people from a town with a population of 14,878 were asked to name their favorite flavor of ice cream. If 7 people in the sample named chocolate as their favorite ice-cream flavor, about how many people in the town would be expected to name chocolate?

- A. 350
- B. 2,100
- C. 7,500
- D. 10,500

Q9

PROBLEM-SOLVING AND DATA ANALYSIS

There are 350 objects in a box. Of these objects, 4% are balls. How many balls are in the box?

- A. 4
- B. 14
- C. 70
- D. 346

-13, 4, 23

A data set of three numbers is shown. If a number from this data set is selected at random, what is the probability of selecting a negative number?

A. 0

B. $\frac{1}{3}$

C. $\frac{2}{3}$

D. 1